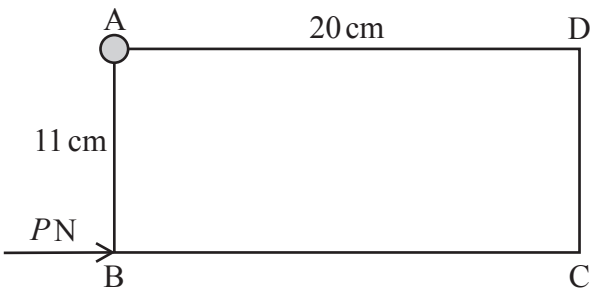
- The acceleration due to gravity is denoted by $g\,\text{m s}^{-2}$. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.

5 A uniform rectangular lamina ABCD has a mass of 0.12 kg. Length AB is 11 cm and length AD is 20 cm.

The lamina ABCD is held by a smooth hinge at A.

A horizontal force of magnitude P N is applied at B so that the lamina is in equilibrium in a vertical plane with AD horizontal as shown in the diagram.



Show that P is 1.07 to **3** significant figures. **[3]**